TRANSFINITTE · HARDWARE TRACK · PRODUCT BRIEF (DESIGN LOCKED)

HridayMitra: a heart check-in station and care loop for Indian homes

A seated 90-second morning check-in, a chest locket, and a care loop that makes sure a person acts on every alert. Built to be sold as a product, not shown as a gadget.

Monitoring only, not a diagnosis tool. It does not replace a doctor or emergency care. Nothing here is built or tested yet.

1. The product in one paragraph

HridayMitra is for people living with heart failure and the family who worry about them. Each morning the patient sits on the station for about 75–90 seconds and wears a small locket day and night. The system learns the patient's own normal. When weight, fluid, heart rate, breathing or sleeping position drift away from it, the device speaks to the patient and sends an explained alert by SMS and voice call to the family, the ASHA worker and the doctor. The alert asks for a reply, and escalates if nobody answers. The aim is to warn several days before a hospital emergency. That aim is untested.

What is in the box

Item	What it is
Check-in station	A foldable all-in-one stool: seat, footrest with 4 foot electrodes, and a grab handle with 4 hand electrodes and a finger pulse and SpO_2 clip. 4 load cells under the frame feet and an electronics box under the seat holding the ESP32, a louder speaker, 3 lights, 2 big buttons and an SOS button, the SIM800L modem, an 18650 battery and a mains charger. A side dock charges the locket.
Locket	A pendant on a cord with a motion sensor, SOS button and small battery. Stores 7 days of data and syncs to the station. Charges in the station's side dock.
Quick-start card	Picture steps and the family, ASHA and doctor contact numbers
ASHA card	What to check and when to call the doctor (the wording must be approved by a cardiologist before use)

Setup, about 10 minutes, by the ASHA worker or a family member: choose the language, enter the hospital's "dry weight", add two family numbers, the ASHA number and the doctor's number, and place a known weight on the station once to calibrate it.

The daily routine

When	What happens	Time
Same time each morning	The station speaks a reminder in the local language. The patient sits, takes off slippers, answers a breathlessness question, rests bare feet on the plates and hands on the handle, answers the pillow question and "did you take your pills?". A light and a short spoken result follow.	75–90 s
Evening, same time daily (after dinner)	A 20-second foot-only reading, compared with the morning one to show how much fluid moved from the legs toward the chest overnight (the fluid-shift index).	~20 s
All day and night	The locket records quietly. SOS and fall detection work at home. Away from home it only logs and syncs later.	0 s

Total about 95–110 seconds a day. Family members can use the station. They are logged as guests and never counted for the patient.

2. What makes it different

Idea	What it is	Why it stands out
1. Fluid and heart-pumping from the same electrodes	The 8 electrodes measure steady impedance (fluid in the legs, separate from the chest) and the tiny pulse in the same signal (a stroke-volume trend).	Most teams measure fluid or heart rate, not both. A scale with foot impedance reported stroke-volume agreement of r=0.81 against the reference method in 32 patients (PMC, company collaboration). **The pulse part is a stretch goal**: the signal is tiny. If it fails, heart rate comes from the finger sensor.
2. Two-witness alerts with confidence	Red needs two independent channels to agree (station and locket) or a repeat reading. Every reading carries a confidence score. A bad reading shows "no reading".	Directly answers false detection.
3. Closed care loop	The alert SMS asks for a reply (1 = I'm checking, 2 = ask the ASHA to visit, 3 = already at the doctor). With no reply, it escalates on a timer.	A large trial of daily telemonitoring cut mortality when a team acted on the data (TIM-HF2). Most designs alert and stop.
4. Fluid-shift index	A 20-second evening foot reading compared with the morning one shows how much leg fluid moved toward the chest overnight. It is tied to the locket's night breathing and lying-flat angle.	The overnight leg-to-chest shift is documented in sleep-apnea research (PMC3607865). **It is a hypothesis for heart failure at home**: I found no study of it. It links legs, lungs and lying flat in one idea and costs ₹0 in hardware.

Supporting features: identity check with guest mode, "silence is never shown as fine" (a missed check-in or dead battery always produces a message), starting from the hospital's dry weight for new patients, cause tags (infection-like, salty meal, after missed medicine, dehydration-like), and a spoken reason with every alert.

3. How each PS sign is measured

PS warning sign	Measured by	Honest status
Weight gain	Load cells	Direct
Faster resting heart rate	Finger sensor, seated at rest	Direct, but a morning reading, not a night one
Faster breathing	Locket at night (rate and shallow-breathing index)	Depends on the locket
Less activity	Locket	Depends on the locket
Fluid in legs	Foot impedance, separate from the chest path	Direct by design, unproven in our circuit
Fluid in lungs	Chest-path impedance, night breathing, lying-flat signals, SpO_2, symptoms	**Estimated, not measured.** No ₹5,000 device can measure lung water directly.
Trouble breathing lying flat	Locket body angle, plus the pillow question	Depends on the locket

Method, and how reliable it is

- **Legs:** impedance at about 5 kHz and 50 kHz, using the classic ESP32's own signal generator (about 130 Hz–55 kHz per Espressif docs, not bench-tested). Low frequency passes mostly through fluid outside the cells, so the ratio of the two readings separates fluid from body size and skin contact. Same time each morning, seated, bare feet, three short readings with the median taken, a reference resistor checked every time, and a contact check that asks the patient to dry their feet. Weight and symptoms confirm a trend before an alert.
- **Lungs:** a congestion index, not one measurement. Night breathing and lying-flat signals carry the most weight, SpO_2 and symptoms add, and chest-path impedance adds least, because the trunk is only about 25 Ω against about 237–310 Ω for a leg or arm (segmental study). A lung alert needs at least two signals to move together.

Evidence, honestly: weight rules alone caught about 23–35% of events. A scale adding foot impedance caught 70% in one company-linked study (SCALE-HF 1). A chest-impedance device found radiographic lung fluid in emergency patients with 90% sensitivity and 94% specificity (one older study). None of this has been tested with our hardware.

4. Alerts and the care loop

- **Personal baseline:** rolling median and spread of each signal. Only worsening counts. Three families: fluid (weight, leg and chest impedance), cardio-respiratory (heart rate, breathing, SpO_2) and symptoms and behaviour.
- **Amber:** one day above the threshold. The device asks for a repeat check.
- **Red:** above threshold on 2 of the last 3 days with at least 2 families, and the two channels agree.
- **Danger rules:** SpO_2 below 90%, "breathless at rest" or SOS act at once.

Level	Patient (device)	Who is told, and when
Green	Green light, "All fine"	Weekly "device OK" SMS to family
Missed check-in	Reminder next time	Family SMS after about 11 am; ASHA SMS after 2 days
Amber	"Please check again this evening"	Family SMS
Red	Red light and spoken reason	Family 1: voice call and SMS. No reply in 2 hours: family 2 and ASHA. No reply in 6 hours: doctor SMS, repeat the calls.
Urgent	Flashing red, "Go to hospital, call 108"	2 family numbers, ASHA and doctor at once

Nothing calls an ambulance automatically. Reply-by-SMS depends on the SIM800L reading incoming messages (confirm on the bench). Timer values are proposals.

5. Cost against the ₹5,000 limit (₹)

Part	₹	Status
Station: foldable stool, 4 load cells and HX711, impedance circuit with switch, finger pulse and SpO_2 sensor, ESP32, louder 4Ω 3W speaker, buttons and LEDs, battery and charger, SIM800L, electronics box	4,032	Prices web-searched 2026-10-24 and accepted. Estimates remain for the foot and hand electrodes, stool frame, buttons and wires.
Locket: ESP32-C3, motion sensor, battery, case	733	Estimate remains for the case
Total	**≈ 4,765**	**Headroom about ₹235**

- **Headroom is thin, and the limit is unproven.** A full itemized component list with every small part (power, protection, board, wiring, mechanical) totals about ₹4,125 at low prices and ₹6,209 at planning prices (see the component list handout). P1's first job is real quotes for every part.
- **ECG is skipped** for now (finger pulse only). If cost trims free about ₹450–530, revisit it, since the hand electrodes could be shared with the impedance circuit. That choice is made last.
- SIM800L is 2G and does not work on Jio. Demo on an Airtel, Vi or BSNL SIM. A 4G modem (about +₹600–800) is the product upgrade and would exceed the limit.
- Left out of the build: chest puck, pill-tray hardware, multi-frequency impedance, voice-based lung analysis. These are future add-ons.

6. The business (all numbers are hypotheses to test)

Question	Answer
Who uses it	An elderly heart failure patient, in a rural or urban home, plus the family and the local ASHA worker
Who pays	Children living in cities; hospitals for the 90 days after discharge; government health programmes and CSR funds
What is sold	The kit at close to parts cost, plus a monthly care-loop service (alerts, the ASHA workflow, a doctor dashboard). Guess: buy at ₹7,000–10,000, or rent at ₹400–500 a month.
Why it's hard to copy	The care loop (who answers, how fast, what worked) and each patient's long-term baseline, not the sensors
Competitors	Dozee (under-mattress, mostly hospitals); Bodyport (US foot-impedance scale, no India layer)
Where it's sold	Hospital discharge desks, pharmacies, online for urban buyers, and the health-centre and ASHA network for rural homes
Expansion	The same weight, fluid and vitals could serve kidney and dialysis patients (my idea, unverified)
Path	1. Hospital pilot with alerts to the study team only. 2. CDSCO registration (probably Class B or C, my estimate; the software guidance was still a draft in October 2025). 3. Online, pharmacy and programme sales.
A cheaper first product	A flat portable scale version (weight, foot impedance, voice, SMS), sold first. It is a roadmap item, not part of the hackathon build.

7. PS requirements and build status

PS requirement or deliverable	Status	Owner
Under 2 minutes a day, no all-day patch	✅ By design	All
Local language, little reading; power cuts; weak internet	✅ By design (2G fails on Jio)	P1, P5
Parts under ₹5,000	⚠️ Unproven: ₹4,765 on the accepted list; ₹4,125–6,209 on the full itemized list	P1
User journey (one page or storyboard)	⚠️ Draft in this brief	P5
Block diagram; schematic of the impedance front end with the ESP32	❌ Not drawn	P1
Simulation proof (LTspice, Wokwi, Python)	❌ Not done	P1, P2

Code: raw signals to alert, personal baseline, a reason for every alert	❌ Not written	P4, P2
Simple screen for patient, family, doctor	❌ Not built	P5
Synthetic data: 20+ patients × 30 days, worsening 3–10 days before the visit	❌ Not generated	P3
Results: sensitivity, specificity, median lead time, false alerts per patient-month, patient-level split	❌ Not run	P4
Heart-rate test on the BIDMC CHF database	❌ Not done	P2
Public GitHub with setup steps and a working demo	❌ Not created	P5
2-page report, 5-minute pitch, "not a diagnosis tool" disclaimer	⚠️ Disclaimer done, rest not written	P5, all

Build order: (1) synthetic patients and the alert engine, which is the core of the score and needs no hardware; (2) the schematic and simulations; (3) the screen, GitHub, report and pitch.

Validation plan

- At least 200 synthetic patients, each with a 30-day warm-up plus a 30-day test window. Worsening starts 3–10 days before the visit, so the median lead time cannot exceed 10 days. 30–40% of events show no weight gain. Model infection, atrial fibrillation, salty meals, medicine changes, missed pills, guest users, wet feet and missing days, and write every assumption down.
- Split by patient, tune on one group and run the test group once. Report sensitivity, specificity, median lead time, false alerts per patient-month, and compare with weight rules (about 23–35% sensitivity). Run ablations: weight only, then each added signal.
- Test the heart-rate code on BIDMC CHF.
- Show a "break it" demo: guest user, wet feet, missed days, infection, power cut.
- Synthetic results only show that the logic finds the patterns we put in. They do not prove clinical accuracy. Say so.

Team

Person	Owns
P1 Hardware	Block diagram, schematics (impedance circuit, locket), LTspice and Wokwi proofs, parts list with real quotes and links
P2 Signals	Pulse and SpO_2 quality, breathing and body angle from the locket, impedance processing, BIDMC test
P3 Synthetic data	200+ patients with all signals and confounders, `DATA.md`
P4 Alert engine	Baselines, scores, two-witness rule, care-loop timers, cause tags, evaluation
P5 Product	Voice prompts in local languages, screens, phone mock-up of the alert sequence, user journey, report, pitch, GitHub

8. Limits and risks

- **No claim of 100% accuracy.** Even implants catch about 70% of events. Lung fluid is estimated.
- **Unproven:** our impedance circuit, handle electrodes giving a separate chest reading, the pulse-impedance signal, the locket's breathing and body-angle quality and its 2–3 week battery life, and reading incoming SMS on the SIM800L.
- **Cost:** some prices are still estimates, the accepted list leaves about ₹235 of headroom, and the full itemized list ranges ₹4,125–6,209.
- **Safety:** impedance current for pacemaker or defibrillator users was not researched. Include an "impedance off" setting for the doctor.
- **It cannot predict sudden death** or replace emergency care. The response loop decides outcomes more than the sensors.

- **Adoption:** large trials saw daily use fall to about 55%. Weekly family messages and missed-day alerts are the main defence.

Key sources: SCALE-HF 1 · thoracic impedance vs chest X-ray · Bodyport stroke volume · TIM-HF2 · HeartLogic · Gyllensten 2016 · Kerala heart failure registry. Several numbers come from abstracts or company-linked studies. Check the full papers before quoting them.